



DEPT. OF MECH. IND. & AERO ENG.

MIAE 221 - Materials Science

Lectures: Wed and Fri Zoom from 08:45 to 10:00 am

Instructor: **Dr. Mamoun Medraj, P.Eng**
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Dept. of Mech. Ind. & Aero. Eng.,
Room, EV 12.185

Textbook:
W.D. Callister, Materials Science & Engineering: An Introduction
5th to 9th ed., J. Wiley.

Handouts and Coursework: Available on the **course Moodle**



MIAE 221 - Materials Science

Assessment:

- In-Tutorial Problems 20 %
- Term-Assignment 10 %
- Midterm 25 %
- Final 45 %

- Assignments will not be collected but the questions will be used as the basis for the *In-Tutorial Assignment Problems*.

- Also, some of the assignment problems and tutorials questions (*or similar ones*) will be asked in the exams. The solutions will be discussed in the tutorial sessions.

The *In-Tutorial Assignment Problems* will take place **every second** tutorial. The first one will be during the week of Monday Sep. 21st

Midterm Exam: **Friday October 30th, 2026**

Midterm and final exams will be done in class. More details will be provided later.



MECH 221 - Materials Science

Tutorials sections:

- **XA:** Friday at 14:45-15:35 in H 564
- **XB:** Friday at 14:45-15:35 in H 562
- **XC:** Monday at 17:45-18:35 in CL 235
- **XD:** Friday at 16:15-17:05 in CL 235
- **XF:** Wednesday 14:45-15:35 in H 561

Teaching Assistants:

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What is Material?

Oils, gases, pharmaceuticals :

Iron, copper, polymers, cement:

- *Material can be put into certain geometric shape*
- *The product has some **functionality***

Old materials: metals, wood, ceramics, skins,
natural fibers, (papyrus)

New materials: **metals, ceramics, natural &
synthetic fibers, polymers**



Introduction: *Historical Perspective*

- **Civilization** strongly linked with **materials**
Stone age, iron age, bronze age... nuclear age, information age
- Sumerians: ceramics
- Egyptians: lime
- Anatolians: Iron (12th century BC)
- The earliest known Bronze is from what is now Iran and Iraq



Introduction

Technological advances have been materials driven:

- **Transportation**; engines, airframes, auto bodies
 - **Space exploration**; shuttle tiles, high temp alloys
 - **Energy**; solar power, batteries
 - **Communications**; semiconductors
-
- Military uses $\Rightarrow$ Commercial uses



What is Materials Science ?

- Relationships between structure and
..... of materials

What is Materials Engineering ?

- Structure-property correlations
- Design the structure of a material to impart some desired properties



Introduction

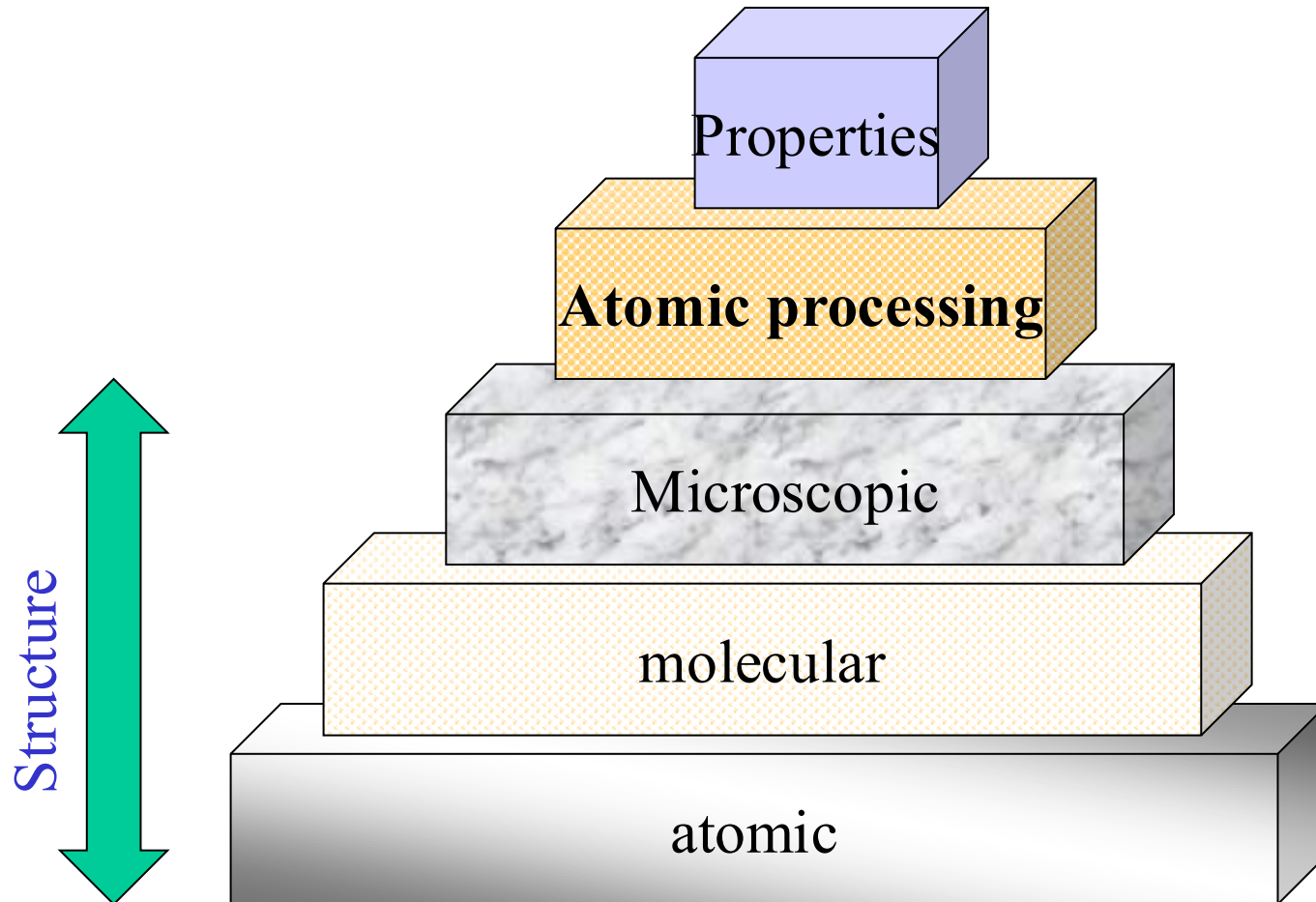
Property: Response of a material to
an effect, such as

- mechanical
- thermal
- electrical
- magnetic
- optical

Properties are independent of material shape & size



General Course Outline:





Course Outline

Topic	Lectures
• Introduction: <i>Basic characteristics of metals, ceramics, polymers and composites, engineering design criteria</i>	1
• Chemistry Review: <i>Atomic structure and Interatomic Bonding</i>	2
• The structure of Crystalline Solids	3
• Imperfections in Crystals	2
• Diffusion	1
• Mechanical Properties of Metals	3
Midterm Exam	
• Phase Diagrams	3
• Structure, Properties, and Applications of Ceramics	1
• Polymer Structures and Applications	1
• Thermal, Electrical, Magnetic and Optical Properties	4

Exam → 1 lecture
Review → 2 lect's
Total: 24 lectures

Σ 21



Why study Materials Science?

(1) Important to understand **capabilities and limitations of materials:**

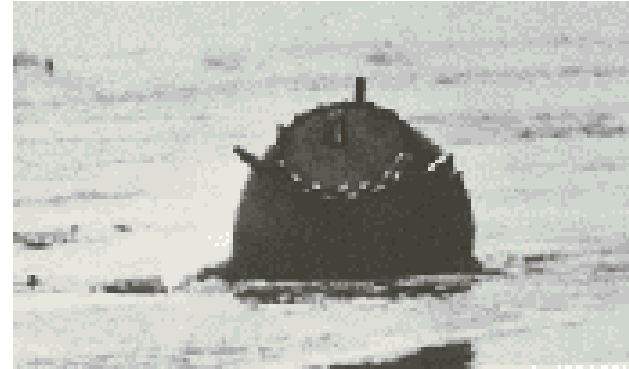
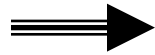
- The following are just a few examples of **catastrophic failure** caused by a lack of fundamental understanding of materials, their properties, and failure modes.



Examples of *Catastrophic Failure*



Liberty ships (WWII)



D-B-T in BCC Fe (metal)



Challenger (1986)



failure of an O-ring seal (polymer)



Examples of *Catastrophic Failure*



Hyatt Regency (KC)
walkway collapse (1981)



Overstressed
steel support
rods
(underdesigned)

Death 114 people



Alaska MD-80 crash (1999)



Excessive
wear on
stabilizer
jackscrew

Death 88 people



Examples of *Catastrophic Failure*

- **Tacoma Narrows Bridge Collapse** (1940)

poor design – **insufficient crosswind stiffening**



- **de Havilland Comet** (first commercial jet) (1954)

metal fatigue, aggravated by high stresses around rivet holes near window openings

- **United DC-10** crash (Sioux City, IA) (1989)

inclusion and cracking in primary #2 engine turbine blade



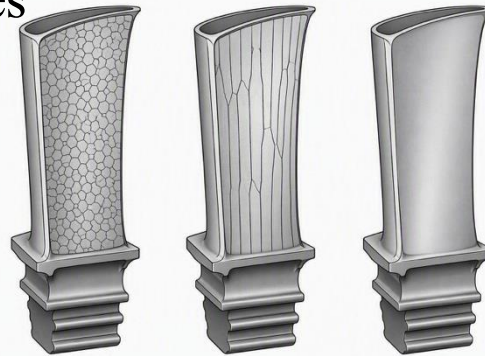
Why Study Materials Science?

(2) An understanding of Materials Science help us to **design better components, parts, devices, etc.**

- how can we make something stronger or lighter?
- how do elements come together to form alloys?
- why do some materials have vastly different properties than others?

Example: jet engine turbine blades

Grain boundaries creep at high temperature → removing them (equiaxed → columnar → single crystal) let designers raise operating temperature by $\sim 200^{\circ}\text{C}$ and improve engine efficiency.



(3) It's *interesting* and helps to make you a more informed person



Next time: **Chemistry Review**